Less is More

Homework 5

1. (a) Both figures plot the training loss of models of different hidden layer width vs the log of the learning rate. Larger width lead to better performance when using the muP scaling.

1.(b): 1

1.(c): ① $\|h_l\|_2 = \Theta(\sqrt{n_l})$ and $\|\Delta h_l\|_2 = \Theta(\sqrt{n_l})$ ensures that $\|h_l\|_{RSM} = \Theta(1)$ and $\|\Delta h_l\|_{RSM} = \Theta(1)$ which prevents the features and their updates from exploding or vanishing.

② $\|h_l\|_{RSM} = \frac{\|h_l\|_{RSM}}{\sqrt{n_l}} = \Theta(1)$, $\|\Delta h_l\|_{RSM} = \frac{\|\Delta h_l\|_{RSM}}{\sqrt{n_l}} = \Theta(1)$

③ $\because h_l = W_l \cdot h_{l-1}$

$\therefore \|h_l\|_2 = \|W_l h_{l-1}\|_2 \le \|W_l\|_2 \|h_{l-1}\|_2 = \Theta(\sqrt{n_l})$

$\therefore \|h_l\|_2 \le O(n_l)$

$\because \Delta h_l + h_l = (\Delta W_l + W_l)(\Delta h_{l-1} + h_{l-1})$

$\therefore \Delta h_l = \Delta W_l \Delta h_{l-1} + \Delta W_l \cdot h_{l-1} + W_l \cdot \Delta h_{l-1} + W_l \cdot h_{l-1} - h_l$

$= \Delta W_l \Delta h_{l-1} + \Delta W_l \cdot h_{l-1} + W_l \cdot \Delta h_{l-1}$

Similarly: $\|\Delta h_l\|_2 \le \|\Delta W_l\|_2 \|\Delta h_{l-1}\|_2 + \|\Delta W_l\|_2 \|h_{l-1}\|_2 + \|W_l\|_2 \|\Delta h_{l-1}\|_2$

$\therefore \|\Delta h_l\|_2 \le \Theta\left(\sqrt{\frac{n_l}{n_{l-1}}}\right) \cdot \|\Delta h_{l-1}\|_2$

I will use mathematical induction to prove the rest:

Base case: Since h_0 will not be changed, therefore $\|\Delta h_0\| = 0$, which stands for $\|\Delta h_0\| = O(\sqrt{n_0})$

Inductive Hypothesis: Assume that for layer $l-1$, $\|\Delta h_{l-1}\| = O(\sqrt{n_{l-1}})$

Inductive steps: $\because \|\Delta h_l\|_2 \le \Theta\left(\sqrt{\frac{n_l}{n_{l-1}}}\right) \cdot \|\Delta h_{l-1}\|_2$

$\therefore \|\Delta h_l\|_2 \le \Theta\left(\sqrt{\frac{n_l}{n_{l-1}}}\right) \cdot O(\sqrt{n_{l-1}})$

$\therefore \|\Delta h_l\|_2 \le \Theta(\sqrt{n_l})$

④ $\|W_l h_{l-1}\| = \Theta(\|W_l\| \cdot \|h_{l-1}\|)$

$\|\Delta W_l h_{l-1}\| = \|\Delta W_l\| \cdot \|h_{l-1}\|$

Less is More

3.(a): Since FP8 has limited numerical precision. the small weights produced by Xavier initialization may go beyond the precision that FP8 can handle. So, $N(0,1)$ may work better.

3.(b): Xavier : $N(0, \frac{1}{\sqrt{d_{in}}})$

$\therefore C = \frac{1}{\sqrt{d_{in}}}$

3.(c): $\Delta y = C \cdot \Delta W \cdot X$

$$\|\Delta y\|_{RMS} = C \cdot \|\Delta W \cdot X\|_{RMS} \leq C \cdot \|\Delta W\|_{RMS-RMS}^{d_{out} \times d_{in}} \cdot \|X\|_{RMS}$$

$$= C \cdot \frac{\|\Delta W \cdot m\|_{RMS}}{\|m\|_{RMS}} \cdot \|X\|_{RMS}$$

$$= C \cdot \sqrt{\frac{d_{in}}{d_{out}}} \cdot \|\Delta W\|_2 \cdot \|X\|_{RMS}$$

$\because \|X\|_{RMS} = 1, \quad C = \frac{1}{\sqrt{d_{in}}}$

$\therefore \|\Delta y\|_{RMS} \leq \frac{1}{\sqrt{d_{in}}} \cdot \sqrt{\frac{d_{in}}{d_{out}}} \cdot \|\Delta W\|_2 \cdot 1 = \frac{1}{\sqrt{d_{out}}} \cdot \|\Delta W\|_2$

$\therefore \|\Delta W\|_2 \leq \sqrt{d_{out}}$

3.(d) $y = C \cdot W \cdot x$ (d_{out})

$\frac{\partial L}{\partial W} = \frac{\partial L}{\partial y} \cdot C \cdot X^T$

$\because$ minibatch $= 1$

$\therefore X^T \in d_{in} \times 1$

$$\begin{pmatrix} y_1' \\ y_2' \\ \vdots \\ y_{out}' \end{pmatrix} \begin{bmatrix} x_1 & x_2 & \cdots & x_{in} \end{bmatrix} = \begin{pmatrix} y_1'x_1 & y_1'x_2 & \cdots & y_1'x_{out} \\ \vdots & & & \vdots \\ y_{out}'x_1 & y_{out}'x_2 & \cdots & y_{out}'x_{out} \end{pmatrix}$$

$\therefore \text{rank}(\text{sign}\,\Delta W L) = 1$

$\therefore \|\text{sign}\,\Delta W L\|_2 = \|\text{sign}\,\Delta W L\|_F = \sqrt{d_{in} \cdot d_{out}}$

$\therefore a \leq \frac{1}{\sqrt{d_{in}}}$

3(e): $\|\Delta W\|_2 = a \cdot \|U\|_2 \cdot \|V^T\|_2$

$= a \cdot 1 \cdot 1$

$= a$

$\therefore a = \sqrt{d_{out}}$

3(g) $\nabla_{X_n} L = C_n \cdot W_n^T \cdot \nabla_{X_{n+1}} L$

$\|C_n \cdot W_n^T\|_2 = C_n \cdot \sqrt{d_{out}} = 1$

$\therefore C_n = \frac{1}{\sqrt{d_{out}}}$

4. (a): $\begin{bmatrix} -40 & -40 & -40 \\ -40 & -40 & -40 \\ -40 & -40 & -40 \end{bmatrix}$

(b): (i): $\begin{bmatrix} -19 & -28 & -32 & -36 & -29 \\ -30 & -40 & -40 & -40 & -30 \\ -30 & -40 & -40 & -40 & -30 \\ -30 & -40 & -40 & -40 & -30 \\ 49 & 68 & 72 & 76 & 59 \end{bmatrix}$

(ii): $\begin{bmatrix} -19 & -32 & -29 \\ -30 & -40 & -30 \\ 49 & 72 & 59 \end{bmatrix}$

5. (a): weights: $C \cdot K^2 \cdot F$, biases: F

Shape: $\left(\left\lfloor \frac{W+2P-K}{S} \right\rfloor\right) \times \left(\left\lfloor \frac{H+2P-K}{S} \right\rfloor\right) \times F$

(b): $\left(\left\lfloor \frac{W-K}{S} \right\rfloor\right) \times \left(\left\lfloor \frac{H-K}{S} \right\rfloor\right) \times C$

(c): $L=1$, $Size = K$

$L=2$, $Size = 2K-1$

$L=3$, $Size = 3K-2$

$\vdots$

$Size = L(k-1)+1$

(d): Let R be the receptive field size before pooling, and let J be the distance between the receptives of two adjacent feature points in the original input.

$\because$ The size of the kernel is 2

$\therefore R_{after} = R + J$

Assume that the size of the item is $M \times N \times C$ before doing the downsampling, after doing the downsampling, the size will turn to $\frac{M}{2} \times \frac{N}{2} \times 2C$.

Therefore, the total amount of the feature values is reduced by half.

(e):

Layer	Number of Parameters	Dimension of Activations
Input	0	$28 \times 28 \times 1$
CONV3-10	100	$28 \times 28 \times 10$
POOL2	0	$14 \times 14 \times 10$
CONV3-10	$3 \times 3 \times 10 \times 10 + 10$	$14 \times 14 \times 10$
POOL2	0	$7 \times 7 \times 10$
FLATTEN	0	490
FC-3	$490 \times 3 + 3$	3

(f): $g_3 = g_2 = \begin{bmatrix} 0 \\ 0.8 \\ 0 \end{bmatrix}$

$g_4 = g_1 = \begin{bmatrix} 0.8 \\ 0 \\ 0 \end{bmatrix}$

7. (a): $H_{out} = \lfloor \frac{H_{in}+2P-K}{S} \rfloor + 1$, $W_{out} = \lfloor \frac{W_{in}+2P-K}{S} \rfloor + 1$

(i): $H_{out} = \lfloor \frac{32+2-3}{1} \rfloor + 1 = 32$, $W_{out} = \lfloor \frac{32+2-3}{1} \rfloor + 1 = 32$, $C_{out} = 64$

∴ Shape will be $[10, 64, 32, 32]$

(ii): $H_{out} = \lfloor \frac{32+0-4}{2} \rfloor + 1 = 15$, $W_{out} = \lfloor \frac{32+0-4}{1} \rfloor + 1 = 15$, $C_{out} = 64$

∴ Shape will be $[10, 64, 32, 32]$

(b): $\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$ (c): $\begin{bmatrix} \frac{1}{9} & \frac{1}{9} & \frac{1}{9} \\ \frac{1}{9} & \frac{1}{9} & \frac{1}{9} \\ \frac{1}{9} & \frac{1}{9} & \frac{1}{9} \end{bmatrix}$

(d): (i): $K' = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

$M * K' = \begin{bmatrix} 1+3+7+9 \end{bmatrix} = \begin{bmatrix} 20 \end{bmatrix}$

(ii): $r = 3 + (5-1) \times 1 = 7$

∴ 7×7 is the receptive field